

Miles Franck

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SUMMARY

Software engineering graduate with a strong foundation in Python, Rust, and database systems. Experienced building automation tools and data pipelines, with applied knowledge of secure coding practices, network systems, and Agile workflows. Eager to contribute to a collaborative engineering team as an entry-level software engineer.

TECHNICAL SKILLS

Languages: Python, Rust, SQL

Databases: PostgreSQL, MySQL

Tools & Platforms: Git, WSL, Nmap, Shodan, Splunk, Wireshark, Palo Alto, Claude Code

Methodologies: Agile / Scrum, Threat Modeling, Secure SDLC

Concepts: Network Architecture (OSI / VLANs), Cryptography, REST APIs, Data Sanitization

PROJECTS

Automated Vulnerability Analysis Tool | Personal Project2024

- Built a Python tool that queries the Shodan API to enumerate internet-facing assets and cross-references results against CVE/CPE vulnerability databases.
- Generated structured attack-path reports to surface and prioritize exploitable risks — translating raw data into actionable output.
- Applied object-oriented design and modular architecture to keep the codebase extensible for future data sources.

Secure File-Deletion Utility | Personal Project2024

- Developed a command-line tool in Python implementing NIST-aligned data sanitization standards for secure file removal.
- Designed with configurable overwrite passes and logging to support auditability and compliance use cases.

Network Asset Discovery Tool | Personal Project2025

- Built an nmap-driven tool to automate network scanning and asset enumeration, returning structured output for downstream analysis.

Personal SQL File Storage System | Personal Project*In Progress*

- Designing a relational database schema in PostgreSQL to index and retrieve personal file metadata.
- Focused on query optimization, schema normalization, and efficient storage patterns.

Academic Grading System — XML Pattern Engine | Allegheny College2022–2025

- Contributed to the college grading platform by engineering XML parsing patterns that evaluate student submissions against professor-defined requirements.
- Collaborated with faculty to refine pattern logic and reduce false positives in automated grading output.

CERTIFICATIONS & TRAINING

CompTIA Security+ | CompTIA*Expected May 2026*

- Industry-standard credential covering network security, cryptography, risk management, and compliance.

SOC Operations Analyst II — Live Operations | BCR Cyber*Dec 2025*

- Completed live-fire cyber range exercises emphasizing real-time incident response and threat containment under simulated production conditions.

SOC Operations Analyst I — Fundamentals | BCR Cyber*Nov 2025*

- Coursework: Security frameworks, Linux & Cisco CLI, network architecture (OSI model, VLANs), cryptography, application security, and SIEM log analysis using Splunk.

EDUCATION

B.S. Software Engineering | Allegheny College*May 2025*

- College Trustee Scholarship Recipient (2021–2025)
- Additional Merit Scholarship Recipient (2022–2024)

Experience

Carnegie Mellon Heinz College Summer Fellowship | Carnegie Mellon *2025*

- Paid Summer Fellowship
- Participated in a hands-on IT and cybersecurity program, tackling real-world technical challenges
- Gained experience in system architecture, security solutions, and project management through collaboration with industry professionals.